

# Week 4 Linear Algebra

By

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## CHAPTER 3

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# Euclidean Vector Spaces

## Vectors Whose Initial Point Is Not at the Origin

It is sometimes necessary to consider vectors whose initial points are not at the origin. If  $\overrightarrow{P_1P_2}$  denotes the vector with initial point  $P_1(x_1, y_1)$  and terminal point  $P_2(x_2, y_2)$ , then the components of this vector are given by the formula

$$\overrightarrow{P_1P_2} = (x_2 - x_1, y_2 - y_1) \quad (4)$$

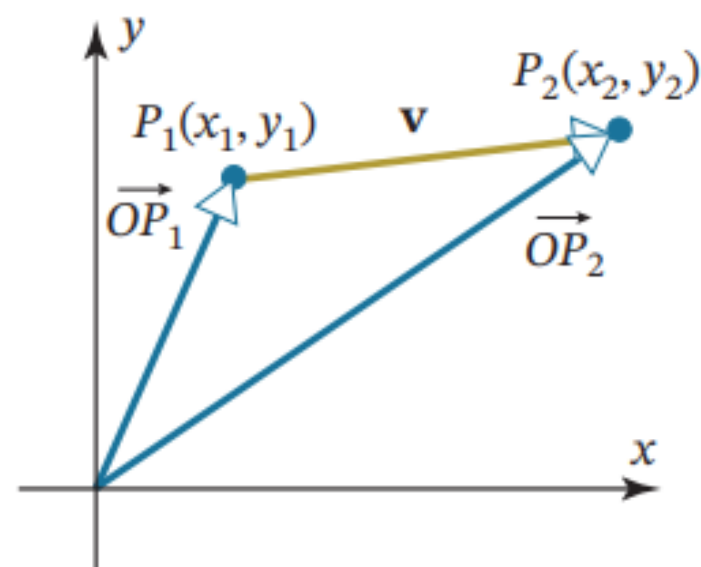
That is, the components of  $\overrightarrow{P_1P_2}$  are obtained by subtracting the coordinates of the initial point from the coordinates of the terminal point. For example, in **Figure 3.1.12** the vector  $\overrightarrow{P_1P_2}$  is the difference of vectors  $\overrightarrow{OP_2}$  and  $\overrightarrow{OP_1}$ , so

$$\overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1} = (x_2, y_2) - (x_1, y_1) = (x_2 - x_1, y_2 - y_1)$$

As you might expect, the components of a vector in 3-space that has initial point  $P_1(x_1, y_1, z_1)$  and terminal point  $P_2(x_2, y_2, z_2)$  are given by

$$\overrightarrow{P_1P_2} = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$

(5)



$$\mathbf{v} = \overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1}$$

**FIGURE 3.1.12**

### EXAMPLE 1 | Finding the Components of a Vector

The components of the vector  $\mathbf{v} = \overrightarrow{P_1P_2}$  with initial point  $P_1(2, -1, 4)$  and terminal point  $P_2(7, 5, -8)$  are

$$\mathbf{v} = (7 - 2, 5 - (-1), (-8) - 4) = (5, 6, -12)$$

# $n$ -Space

## Definition 1

If  $n$  is a positive integer, then an **ordered  $n$ -tuple** is a sequence of  $n$  real numbers  $(v_1, v_2, \dots, v_n)$ . The set of all ordered  $n$ -tuples is called **real  $n$ -space** and is denoted by  $R^n$ .

## Operations on Vectors in $R^n$

Our next goal is to define useful operations on vectors in  $R^n$ . These operations will all be natural extensions of the familiar operations on vectors in  $R^2$  and  $R^3$ . We will denote a vector  $\mathbf{v}$  in  $R^n$  using the notation

$$\mathbf{v} = (v_1, v_2, \dots, v_n)$$

and we will call  $\mathbf{0} = (0, 0, \dots, 0)$  the **zero vector**.

We noted earlier that in  $R^2$  and  $R^3$  two vectors are equivalent (equal) if and only if their corresponding components are the same. Thus, we make the following definition.

### Definition 2

Vectors  $\mathbf{v} = (v_1, v_2, \dots, v_n)$  and  $\mathbf{w} = (w_1, w_2, \dots, w_n)$  in  $R^n$  are said to be **equal** (also called **equivalent**) if

$$v_1 = w_1, \quad v_2 = w_2, \dots, \quad v_n = w_n$$

We indicate this by writing  $\mathbf{v} = \mathbf{w}$ .

## EXAMPLE 2 | Equality of Vectors

The vectors

$$\mathbf{v} = (a, b, c, d) \quad \text{and} \quad \mathbf{w} = (1, -4, 2, 7)$$

are equal if and only if  $a = 1$ ,  $b = -4$ ,  $c = 2$ , and  $d = 7$ .



### Definition 3

If  $\mathbf{v} = (v_1, v_2, \dots, v_n)$  and  $\mathbf{w} = (w_1, w_2, \dots, w_n)$  are vectors in  $R^n$ , and if  $k$  is any scalar, then we define

$$\mathbf{v} + \mathbf{w} = (v_1 + w_1, v_2 + w_2, \dots, v_n + w_n) \quad (10)$$

$$k\mathbf{v} = (kv_1, kv_2, \dots, kv_n) \quad (11)$$

$$-\mathbf{v} = (-v_1, -v_2, \dots, -v_n) \quad (12)$$

$$\mathbf{w} - \mathbf{v} = \mathbf{w} + (-\mathbf{v}) = (w_1 - v_1, w_2 - v_2, \dots, w_n - v_n) \quad (13)$$

### EXAMPLE 3 | Algebraic Operations Using Components

If  $\mathbf{v} = (1, -3, 2)$  and  $\mathbf{w} = (4, 2, 1)$ , then

$$\begin{aligned} \mathbf{v} + \mathbf{w} &= (5, -1, 3), & 2\mathbf{v} &= (2, -6, 4) \\ -\mathbf{w} &= (-4, -2, -1), & \mathbf{v} - \mathbf{w} &= \mathbf{v} + (-\mathbf{w}) = (-3, -5, 1) \end{aligned}$$

### Theorem 3.1.1

If  $\mathbf{u}$ ,  $\mathbf{v}$ , and  $\mathbf{w}$  are vectors in  $R^n$ , and if  $k$  and  $m$  are scalars, then:

(a)  $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$

(b)  $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$

(c)  $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$

(d)  $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$

(e)  $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

(f)  $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

(g)  $k(m\mathbf{u}) = (km)\mathbf{u}$

(h)  $1\mathbf{u} = \mathbf{u}$

## Linear Combinations

Addition, subtraction, and scalar multiplication are frequently used in combination to form new vectors. For example, if  $\mathbf{v}_1$ ,  $\mathbf{v}_2$ , and  $\mathbf{v}_3$  are vectors in  $R^n$ , then the vectors

$$\mathbf{u} = 2\mathbf{v}_1 + 3\mathbf{v}_2 + \mathbf{v}_3 \quad \text{and} \quad \mathbf{w} = 7\mathbf{v}_1 - 6\mathbf{v}_2 + 8\mathbf{v}_3$$

are formed in this way. In general, we make the following definition.

### Definition 4

If  $\mathbf{w}$  is a vector in  $R^n$ , then  $\mathbf{w}$  is said to be a **linear combination** of the vectors  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$  in  $R^n$  if it can be expressed in the form

$$\mathbf{w} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \cdots + k_r\mathbf{v}_r \tag{14}$$

where  $k_1, k_2, \dots, k_r$  are scalars. These scalars are called the **coefficients** of the linear combination. In the case where  $r = 1$ , Formula (14) becomes  $\mathbf{w} = k_1\mathbf{v}_1$ , so that a linear combination of a single vector is just a scalar multiple of that vector.

In Exercises 3–4, find the components of the vector  $\overrightarrow{P_1P_2}$ .

3. a.  $P_1(3, 5), P_2(2, 8)$

b.  $P_1(5, -2, 1), P_2(2, 4, 2)$

4. a.  $P_1(-6, 2), P_2(-4, -1)$

b.  $P_1(0, 0, 0), P_2(-1, 6, 1)$

### Exercise Set 3.1

- 9.** Let  $\mathbf{u} = (4, -1)$ ,  $\mathbf{v} = (0, 5)$ , and  $\mathbf{w} = (-3, -3)$ . Find the components of
- a.**  $\mathbf{u} + \mathbf{w}$
  - b.**  $\mathbf{v} - 3\mathbf{u}$
  - c.**  $2(\mathbf{u} - 5\mathbf{w})$
  - d.**  $3\mathbf{v} - 2(\mathbf{u} + 2\mathbf{w})$
- 10.** Let  $\mathbf{u} = (-3, 1, 2)$ ,  $\mathbf{v} = (4, 0, -8)$ , and  $\mathbf{w} = (6, -1, -4)$ . Find the components of
- a.**  $\mathbf{v} - \mathbf{w}$
  - b.**  $6\mathbf{u} + 2\mathbf{v}$
  - c.**  $-3(\mathbf{v} - 8\mathbf{w})$
  - d.**  $(2\mathbf{u} - 7\mathbf{w}) - (8\mathbf{v} + \mathbf{u})$
- 11.** Let  $\mathbf{u} = (-3, 2, 1, 0)$ ,  $\mathbf{v} = (4, 7, -3, 2)$ , and  $\mathbf{w} = (5, -2, 8, 1)$ . Find the components of
- a.**  $\mathbf{v} - \mathbf{w}$
  - b.**  $-\mathbf{u} + (\mathbf{v} - 4\mathbf{w})$
  - c.**  $6(\mathbf{u} - 3\mathbf{v})$
  - d.**  $(6\mathbf{v} - \mathbf{w}) - (4\mathbf{u} + \mathbf{v})$
- 12.** Let  $\mathbf{u} = (1, 2, -3, 5, 0)$ ,  $\mathbf{v} = (0, 4, -1, 1, 2)$ , and  $\mathbf{w} = (7, 1, -4, -2, 3)$ . Find the components of
- a.**  $\mathbf{v} + \mathbf{w}$
  - b.**  $3(2\mathbf{u} - \mathbf{v})$
  - c.**  $(3\mathbf{u} - \mathbf{v}) - (2\mathbf{u} + 4\mathbf{w})$
  - d.**  $\frac{1}{2}(\mathbf{w} - 5\mathbf{v} + 2\mathbf{u}) + \mathbf{v}$

## 3.2 Norm, Dot Product, and Distance in $R^n$

### Norm of a Vector

In this text we will denote the length of a vector  $\mathbf{v}$  by the symbol  $\|\mathbf{v}\|$ . As suggested in **Figure 3.2.1a**, it follows from the Theorem of Pythagoras that the norm of a vector  $(v_1, v_2)$  in  $R^2$  is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2} \quad (1)$$

Similarly, for a vector  $(v_1, v_2, v_3)$  in  $R^3$ , it follows from **Figure 3.2.1b** and two applications of the Theorem of Pythagoras that

$$\|\mathbf{v}\|^2 = (OR)^2 + (RP)^2 = (OQ)^2 + (QR)^2 + (RP)^2 = v_1^2 + v_2^2 + v_3^2$$

and hence that

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2} \quad (2)$$

Motivated by the pattern of Formulas (1) and (2), we make the following definition.

### Definition 1

If  $\mathbf{v} = (v_1, v_2, \dots, v_n)$  is a vector in  $R^n$ , then the **norm** of  $\mathbf{v}$  (also called the **length** of  $\mathbf{v}$  or the **magnitude** of  $\mathbf{v}$ ) is denoted by  $\|\mathbf{v}\|$ , and is defined by the formula

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2} \quad (3)$$

### EXAMPLE 1 | Calculating Norms

It follows from Formula (2) that the norm of the vector  $\mathbf{v} = (-3, 2, 1)$  in  $R^3$  is

$$\|\mathbf{v}\| = \sqrt{(-3)^2 + 2^2 + 1^2} = \sqrt{14}$$

and it follows from Formula (3) that the norm of the vector  $\mathbf{v} = (2, -1, 3, -5)$  in  $R^4$  is

$$\|\mathbf{v}\| = \sqrt{2^2 + (-1)^2 + 3^2 + (-5)^2} = \sqrt{39}$$

## Distance in $R^n$

If  $P_1$  and  $P_2$  are points in  $R^2$  or  $R^3$ , then the length of the vector  $\overrightarrow{P_1P_2}$  is equal to the distance  $d$  between the two points (**Figure 3.2.3**). Specifically, if  $P_1(x_1, y_1)$  and  $P_2(x_2, y_2)$  are points in  $R^2$ , then Formula (4) of Section 3.1 implies that

$$d = \|\overrightarrow{P_1P_2}\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (9)$$

This is the familiar distance formula from analytic geometry. Similarly, the distance between the points  $P_1(x_1, y_1, z_1)$  and  $P_2(x_2, y_2, z_2)$  in 3-space is

$$d(\mathbf{u}, \mathbf{v}) = \|\overrightarrow{P_1P_2}\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \quad (10)$$

Motivated by Formulas (9) and (10), we make the following definition.

### Definition 2

If  $\mathbf{u} = (u_1, u_2, \dots, u_n)$  and  $\mathbf{v} = (v_1, v_2, \dots, v_n)$  are points in  $R^n$ , then we denote the **distance** between  $\mathbf{u}$  and  $\mathbf{v}$  by  $d(\mathbf{u}, \mathbf{v})$  and define it to be

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \cdots + (u_n - v_n)^2} \quad (11)$$



### EXAMPLE 4 | Calculating Distance in $R^n$

If

$$\mathbf{u} = (1, 3, -2, 7) \quad \text{and} \quad \mathbf{v} = (0, 7, 2, 2)$$

then the distance between  $\mathbf{u}$  and  $\mathbf{v}$  is

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{(1 - 0)^2 + (3 - 7)^2 + (-2 - 2)^2 + (7 - 2)^2} = \sqrt{58}$$

## EXAMPLE 6 | Calculating Dot Products Using Components

- (a) Use Formula (15) to compute the dot product of the vectors  $\mathbf{u}$  and  $\mathbf{v}$  in Example 5.  
(b) Calculate  $\mathbf{u} \cdot \mathbf{v}$  for the following vectors in  $R^4$ :

$$\mathbf{u} = (-1, 3, 5, 7), \quad \mathbf{v} = (-3, -4, 1, 0)$$

**Solution (a)** The component forms of the vectors are  $\mathbf{u} = (0, 0, 1)$  and  $\mathbf{v} = (0, 2, 2)$ . Thus,

$$\mathbf{u} \cdot \mathbf{v} = (0)(0) + (0)(2) + (1)(2) = 2$$

which agrees with the result obtained geometrically in Example 5.

**Solution (b)**

$$\mathbf{u} \cdot \mathbf{v} = (-1)(-3) + (3)(-4) + (5)(1) + (7)(0) = -4$$

## EXAMPLE 7 | A Geometry Problem Solved Using Dot Product

Find the angle between a diagonal of a cube and one of its edges.

**Solution** Let  $k$  be the length of an edge and introduce a coordinate system as shown in **Figure 3.2.7**. If we let  $\mathbf{u}_1 = (k, 0, 0)$ ,  $\mathbf{u}_2 = (0, k, 0)$ , and  $\mathbf{u}_3 = (0, 0, k)$ , then the vector

$$\mathbf{d} = (k, k, k) = \mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3$$

is a diagonal of the cube. It follows from Formula (13) that the angle  $\theta$  between  $\mathbf{d}$  and the edge  $\mathbf{u}_1$  satisfies

$$\cos \theta = \frac{\mathbf{u}_1 \cdot \mathbf{d}}{\|\mathbf{u}_1\| \|\mathbf{d}\|} = \frac{k^2}{(k)(\sqrt{3}k)} = \frac{1}{\sqrt{3}}$$

With the help of a calculator we obtain

$$\theta = \cos^{-1} \left( \frac{1}{\sqrt{3}} \right) \approx 54.74^\circ$$

# Algebraic Properties of the Dot Product

In the special case where  $\mathbf{u} = \mathbf{v}$  in Definition 4, we obtain the relationship

$$\mathbf{v} \cdot \mathbf{v} = v_1^2 + v_2^2 + \cdots + v_n^2 = \|\mathbf{v}\|^2 \quad (18)$$

This yields the following formula for expressing the length of a vector in terms of a dot product:

$$\boxed{\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}} \quad (19)$$

Dot products have many of the same algebraic properties as products of real numbers.

## Theorem 3.2.2

If  $\mathbf{u}, \mathbf{v}$ , and  $\mathbf{w}$  are vectors in  $R^n$ , and if  $k$  is a scalar, then:

- (a)  $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$  [Symmetry property]
- (b)  $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$  [Distributive property]
- (c)  $k(\mathbf{u} \cdot \mathbf{v}) = (k\mathbf{u}) \cdot \mathbf{v}$  [Homogeneity property]
- (d)  $\mathbf{v} \cdot \mathbf{v} \geq 0$  and  $\mathbf{v} \cdot \mathbf{v} = 0$  if and only if  $\mathbf{v} = \mathbf{0}$  [Positivity property]

### EXAMPLE 8 | Calculating with Dot Products

$$\begin{aligned}(\mathbf{u} - 2\mathbf{v}) \cdot (3\mathbf{u} + 4\mathbf{v}) &= \mathbf{u} \cdot (3\mathbf{u} + 4\mathbf{v}) - 2\mathbf{v} \cdot (3\mathbf{u} + 4\mathbf{v}) \\&= 3(\mathbf{u} \cdot \mathbf{u}) + 4(\mathbf{u} \cdot \mathbf{v}) - 6(\mathbf{v} \cdot \mathbf{u}) - 8(\mathbf{v} \cdot \mathbf{v}) \\&= 3\|\mathbf{u}\|^2 - 2(\mathbf{u} \cdot \mathbf{v}) - 8\|\mathbf{v}\|^2\end{aligned}$$

## Cauchy–Schwarz Inequality and Angles in $R^n$

Our next objective is to extend to  $R^n$  the notion of “angle” between nonzero vectors  $\mathbf{u}$  and  $\mathbf{v}$ . We will do this by starting with the formula

$$\theta = \cos^{-1} \left( \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \right) \quad (20)$$

which follows from Formula (13) that we previously derived for nonzero vectors in  $R^2$  and  $R^3$ . Since dot products and norms have been defined for vectors in  $R^n$ , it would seem that this formula has all the ingredients to serve as a *definition* of the angle  $\theta$  between two vectors,  $\mathbf{u}$  and  $\mathbf{v}$ , in  $R^n$ . However, there is a fly in the ointment, the problem being that this formula is not valid unless its argument satisfies the inequalities

$$-1 \leq \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \leq 1 \quad (21)$$

Fortunately, these inequalities *do* hold for all nonzero vectors in  $R^n$  as a result of the following fundamental result known as the ***Cauchy–Schwarz inequality***.

### Theorem 3.2.4

#### Cauchy–Schwarz Inequality

If  $\mathbf{u} = (u_1, u_2, \dots, u_n)$  and  $\mathbf{v} = (v_1, v_2, \dots, v_n)$  are vectors in  $R^n$ , then

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\| \quad (22)$$

or in terms of components

$$|u_1v_1 + u_2v_2 + \cdots + u_nv_n| \leq (u_1^2 + u_2^2 + \cdots + u_n^2)^{1/2}(v_1^2 + v_2^2 + \cdots + v_n^2)^{1/2} \quad (23)$$

*In Exercises 9–10, find  $\mathbf{u} \cdot \mathbf{v}$ ,  $\mathbf{u} \cdot \mathbf{u}$ , and  $\mathbf{v} \cdot \mathbf{v}$ .*

**9. a.**  $\mathbf{u} = (3, 1, 4), \mathbf{v} = (2, 2, -4)$

**b.**  $\mathbf{u} = (1, 1, 4, 6), \mathbf{v} = (2, -2, 3, -2)$

**10. a.**  $\mathbf{u} = (1, 1, -2, 3), \mathbf{v} = (-1, 0, 5, 1)$

**b.**  $\mathbf{u} = (2, -1, 1, 0, -2), \mathbf{v} = (1, 2, 2, 2, 1)$

## Cross Product of Vectors

## Definition 1

If  $\mathbf{u} = (u_1, u_2, u_3)$  and  $\mathbf{v} = (v_1, v_2, v_3)$  are vectors in 3-space, then the **cross product**  $\mathbf{u} \times \mathbf{v}$  is the vector defined by

$$\mathbf{u} \times \mathbf{v} = (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1)$$

or, in determinant notation,

$$\mathbf{u} \times \mathbf{v} = \left( \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix}, -\begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix}, \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \right) \quad (1)$$



## EXAMPLE 1 | Calculating a Cross Product

Find  $\mathbf{u} \times \mathbf{v}$ , where  $\mathbf{u} = (1, 2, -2)$  and  $\mathbf{v} = (3, 0, 1)$ .

**Solution** From either (1) or the mnemonic in the preceding remark, we have

$$\begin{aligned}\mathbf{u} \times \mathbf{v} &= \left( \begin{vmatrix} 2 & -2 \\ 0 & 1 \end{vmatrix}, -\begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix}, \begin{vmatrix} 1 & 2 \\ 3 & 0 \end{vmatrix} \right) \\ &= (2, -7, -6)\end{aligned}$$

## Theorem 3.5.1

### Relationships Involving Cross Product and Dot Product

If  $\mathbf{u}$ ,  $\mathbf{v}$ , and  $\mathbf{w}$  are vectors in 3-space, then

$$(a) \quad \mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = 0 \quad [\mathbf{u} \times \mathbf{v} \text{ is orthogonal to } \mathbf{u}]$$

$$(b) \quad \mathbf{v} \cdot (\mathbf{u} \times \mathbf{v}) = 0 \quad [\mathbf{u} \times \mathbf{v} \text{ is orthogonal to } \mathbf{v}]$$

$$(c) \quad \|\mathbf{u} \times \mathbf{v}\|^2 = \|\mathbf{u}\|^2 \|\mathbf{v}\|^2 - (\mathbf{u} \cdot \mathbf{v})^2 \quad [\text{Lagrange's identity}]$$

$$(d) \quad \mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w} \quad [\text{vector triple product}]$$

$$(e) \quad (\mathbf{u} \times \mathbf{v}) \times \mathbf{w} = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{v} \cdot \mathbf{w})\mathbf{u} \quad [\text{vector triple product}]$$

## EXAMPLE 2 | $\mathbf{u} \times \mathbf{v}$ Is Perpendicular to $\mathbf{u}$ and to $\mathbf{v}$

Consider the vectors

$$\mathbf{u} = (1, 2, -2) \quad \text{and} \quad \mathbf{v} = (3, 0, 1)$$

In Example 1 we showed that

$$\mathbf{u} \times \mathbf{v} = (2, -7, -6)$$

Since

$$\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = (1)(2) + (2)(-7) + (-2)(-6) = 0$$

and

$$\mathbf{v} \cdot (\mathbf{u} \times \mathbf{v}) = (3)(2) + (0)(-7) + (1)(-6) = 0$$

$\mathbf{u} \times \mathbf{v}$  is orthogonal to both  $\mathbf{u}$  and  $\mathbf{v}$ , as guaranteed by Theorem 3.5.1.

## Properties of Cross Product

If  $\mathbf{u}$ ,  $\mathbf{v}$ , and  $\mathbf{w}$  are any vectors in 3-space and  $k$  is any scalar, then:

$$(a) \quad \mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})$$

$$(b) \quad \mathbf{u} \times (\mathbf{v} + \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) + (\mathbf{u} \times \mathbf{w})$$

$$(c) \quad (\mathbf{u} + \mathbf{v}) \times \mathbf{w} = (\mathbf{u} \times \mathbf{w}) + (\mathbf{v} \times \mathbf{w})$$

$$(d) \quad k(\mathbf{u} \times \mathbf{v}) = (k\mathbf{u}) \times \mathbf{v} = \mathbf{u} \times (k\mathbf{v})$$

$$(e) \quad \mathbf{u} \times \mathbf{0} = \mathbf{0} \times \mathbf{u} = \mathbf{0}$$

$$(f) \quad \mathbf{u} \times \mathbf{u} = \mathbf{0}$$

$$\mathbf{i} \times \mathbf{i} = \mathbf{0}$$

$$\mathbf{j} \times \mathbf{j} = \mathbf{0}$$

$$\mathbf{k} \times \mathbf{k} = \mathbf{0}$$

$$\mathbf{i} \times \mathbf{j} = \mathbf{k}$$

$$\mathbf{j} \times \mathbf{k} = \mathbf{i}$$

$$\mathbf{k} \times \mathbf{i} = \mathbf{j}$$

$$\mathbf{j} \times \mathbf{i} = -\mathbf{k}$$

$$\mathbf{k} \times \mathbf{j} = -\mathbf{i}$$

$$\mathbf{i} \times \mathbf{k} = -\mathbf{j}$$

## Determinant Form of Cross Product

It is also worth noting that a cross product can be represented symbolically in the form

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k} \quad (4)$$

For example, if  $\mathbf{u} = (1, 2, -2)$  and  $\mathbf{v} = (3, 0, 1)$ , then

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -2 \\ 3 & 0 & 1 \end{vmatrix} = 2\mathbf{i} - 7\mathbf{j} - 6\mathbf{k}$$

## Definition 2

If  $\mathbf{u}$ ,  $\mathbf{v}$ , and  $\mathbf{w}$  are vectors in 3-space, then

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$$

is called the *scalar triple product* of  $\mathbf{u}$ ,  $\mathbf{v}$ , and  $\mathbf{w}$ .

The scalar triple product of  $\mathbf{u} = (u_1, u_2, u_3)$ ,  $\mathbf{v} = (v_1, v_2, v_3)$ , and  $\mathbf{w} = (w_1, w_2, w_3)$  can be calculated from the formula

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} \quad (7)$$

This follows from Formula (4) since

$$\begin{aligned} \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) &= \mathbf{u} \cdot \left( \begin{vmatrix} v_2 & v_3 \\ w_2 & w_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} v_1 & v_3 \\ w_1 & w_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} v_1 & v_2 \\ w_1 & w_2 \end{vmatrix} \mathbf{k} \right) \\ &= \begin{vmatrix} v_2 & v_3 \\ w_2 & w_3 \end{vmatrix} u_1 - \begin{vmatrix} v_1 & v_3 \\ w_1 & w_3 \end{vmatrix} u_2 + \begin{vmatrix} v_1 & v_2 \\ w_1 & w_2 \end{vmatrix} u_3 \\ &= \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} \end{aligned}$$

## EXAMPLE 5 | Calculating a Scalar Triple Product

Calculate the scalar triple product  $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$  of the vectors

$$\mathbf{u} = 3\mathbf{i} - 2\mathbf{j} - 5\mathbf{k}, \quad \mathbf{v} = \mathbf{i} + 4\mathbf{j} - 4\mathbf{k}, \quad \mathbf{w} = 3\mathbf{j} + 2\mathbf{k}$$

**Solution** From (7),

$$\begin{aligned}\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) &= \begin{vmatrix} 3 & -2 & -5 \\ 1 & 4 & -4 \\ 0 & 3 & 2 \end{vmatrix} \\ &= 3 \begin{vmatrix} 4 & -4 \\ 3 & 2 \end{vmatrix} - (-2) \begin{vmatrix} 1 & -4 \\ 0 & 2 \end{vmatrix} + (-5) \begin{vmatrix} 1 & 4 \\ 0 & 3 \end{vmatrix} \\ &= 60 + 4 - 15 = 49\end{aligned}$$



### Theorem 3.5.4

(a) The absolute value of the determinant

$$\det \begin{bmatrix} u_1 & u_2 \\ v_1 & v_2 \end{bmatrix}$$

is equal to the area of the parallelogram in 2-space determined by the vectors  $\mathbf{u} = (u_1, u_2)$  and  $\mathbf{v} = (v_1, v_2)$ . (See **Figure 3.5.7a.**)

(b) The absolute value of the determinant

$$\det \begin{bmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix}$$

is equal to the volume of the parallelepiped in 3-space determined by the vectors  $\mathbf{u} = (u_1, u_2, u_3)$ ,  $\mathbf{v} = (v_1, v_2, v_3)$ , and  $\mathbf{w} = (w_1, w_2, w_3)$ . (See **Figure 3.5.7b.**)

### Theorem 3.5.5

If the vectors  $\mathbf{u} = (u_1, u_2, u_3)$ ,  $\mathbf{v} = (v_1, v_2, v_3)$ , and  $\mathbf{w} = (w_1, w_2, w_3)$  have the same initial point, then they lie in the same plane if and only if

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = 0$$

## Exercise Set 3.5

In Exercises 1–2, let  $\mathbf{u} = (3, 2, -1)$ ,  $\mathbf{v} = (0, 2, -3)$ , and  $\mathbf{w} = (2, 6, 7)$ . Compute the indicated vectors.

- |    |                                                      |                                      |                                                                     |
|----|------------------------------------------------------|--------------------------------------|---------------------------------------------------------------------|
| 1. | a. $\mathbf{v} \times \mathbf{w}$                    | b. $\mathbf{w} \times \mathbf{v}$    | c. $(\mathbf{u} + \mathbf{v}) \times \mathbf{w}$                    |
|    | d. $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w})$ | e. $\mathbf{v} \times \mathbf{v}$    | f. $(\mathbf{u} - 3\mathbf{w}) \times (\mathbf{u} - 3\mathbf{w})$   |
| 2. | a. $\mathbf{u} \times \mathbf{v}$                    | b. $-(\mathbf{u} \times \mathbf{v})$ | c. $\mathbf{u} \times (\mathbf{v} + \mathbf{w})$                    |
|    | d. $\mathbf{w} \cdot (\mathbf{w} \times \mathbf{v})$ | e. $\mathbf{w} \times \mathbf{w}$    | f. $(7\mathbf{v} - 3\mathbf{u}) \times (7\mathbf{v} - 3\mathbf{u})$ |

*In Exercises 13–14, find the area of the triangle with the given vertices.*

**13.**  $A(2, 0), B(3, 4), C(-1, 2)$

**14.**  $A(1, 1), B(2, 2), C(3, -3)$

*In Exercises 15–16, find the area of the triangle in 3-space that has the given vertices.*

**15.**  $P_1(2, 6, -1), P_2(1, 1, 1), P_3(4, 6, 2)$

**16.**  $P(1, -1, 2), Q(0, 3, 4), R(6, 1, 8)$

*In Exercises 17–18, find the volume of the parallelepiped with sides  $\mathbf{u}$ ,  $\mathbf{v}$ , and  $\mathbf{w}$ .*

**17.**  $\mathbf{u} = (2, -6, 2), \mathbf{v} = (0, 4, -2), \mathbf{w} = (2, 2, -4)$

**18.**  $\mathbf{u} = (3, 1, 2), \mathbf{v} = (4, 5, 1), \mathbf{w} = (1, 2, 4)$

**Thank You**